

Application 1

Stores	N	P23	P24
1	20	50	45
2	30	120	100
3	40	250	280
4	10	310	350
5	80	180	220

$$I_{23} = 1300 \Rightarrow I_{24} = ?$$

$$I = \frac{P_1 + P_2 + P_3 + P_4 + P_5}{d}$$

$$I_{23} = \frac{50 + 120 + 250 + 310 + 180}{d} = 1300$$

$$\frac{910}{d} = 1300$$

$$\Rightarrow d = \frac{910}{1300} \approx 0.7$$

$$I_{24} = \frac{45 + 100 + 280 + 350 + 220}{0.7} = 1421.43$$

Relative change:

$$\frac{1421.43 - 1300}{1300} = 9.34\%$$

Event

2024

Drop Stock J $\leftarrow$ 220

Add Stock G $\leftarrow$ 75

Pre-event

$$I_{24} = 1421.43$$

Post-event

$$I_{24} = \frac{45 + 100 + 280 + 350 + 75}{d} \quad ||$$

$$\frac{11}{1421.43}$$

$$\frac{850}{d} = 1421.43$$

$$\Rightarrow d = \frac{850}{1421.43} \approx 0.6$$

Application 2

Event

Stock 3 : 5-for-1 Split

Price changes : $250 \rightarrow \frac{250}{5} = 50$

Pre-event

$$I_{24} = 1450$$

Post-event

$$I_{24} = \frac{60 + 120 + \cancel{50} + 310 + 160}{d} \quad \parallel$$

1450

$$\frac{700}{d} = 1450$$

$$d = \frac{700}{1450} \approx 0.48$$

$$I_{25} = \frac{70 + 100 + 46 + 330 + 150}{0.48} = \dots$$

App 3

Stocks	N	P24	P25
A	25	60	58
B	35	130	95
C	45	80	105
D	15	37	48

value-weighted

$$I = \frac{N_A \cdot P_A + N_B \cdot P_B + N_C \cdot P_C + N_D \cdot P_D}{d}$$

$$I_{24} = \frac{25 \cdot 60 + 35 \cdot 130 + 45 \cdot 80 + 15 \cdot 37}{d} \quad \parallel \quad 1000$$

$$\frac{10205}{d} = 1000$$

$$d = \frac{10205}{1000} = 10.2$$

$$I_{25} = \frac{25.58 + \dots + 15.48}{10.2} = \dots$$